

For loops

Name: _____ Date: _____

Work on paper without running code. For question 5, assume a 300 x 200 canvas.

1. Trace the output

```
for (int i = 2; i < 9; i += 3) {  
    println(i);  
}
```

Write the printed values in order. How many times does the body run?

2. Check the boundary

```
for (int i = 0; i <= 5; i++) {  
    println("Hello");  
}
```

The programmer wants exactly five messages. How many are printed?
Change only the condition to produce exactly five.

3. Can the body be skipped?

```
for (int n = 7; n < 4; n++) {  
    println(n);  
}
```

What is printed? Explain why.

4. Write a loop

Write a for loop that prints 12, 9, 6, 3, 0, in that order.

5. A loop inside draw()

draw() clears the background, then runs this loop each frame:

```
for (int x = 40; x <= 160; x += 40) {  
    ellipse(x, 100, 20, 20);  
}
```

How many circles appear in each frame: one or four? Explain.
